

Shivam Nadimpalli

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POSITIONS	Massachusetts Institute of Technology C.L.E. Moore Instructor of Mathematics (September 2024 to Present) Simons Institute for the Theory of Computing Simons–Berkeley Research Fellow (May 2024 to August 2024)	
EDUCATION	Columbia University PhD in Computer Science (August 2019 to May 2024) Advisors: Rocco A. Servedio and Mihalis Yannakakis Brown University ScB in Mathematics–Computer Science with Honors (September 2015 to May 2019) Advisor: Sorin Istrail	
RESEARCH	<i>Journal Publications</i> 1. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Convex Influences”. <i>Discrete Analysis</i> (2026). arXiv: 2109.03107. To appear. 2. Josh Alman, Shivam Nadimpalli, Shyamal Patel, and Rocco A. Servedio. “Faster exact learning of k-term DNFs with membership and equivalence queries”. <i>Journal of the ACM</i> (2026). To appear. 3. Shivam Nadimpalli and Caleb Pascale. “On the Maximal Gaussian Perimeter of Convex Sets, Revisited”. <i>Geometric Aspects of Functional Analysis</i>. Lecture Notes in Mathematics. 2026. arXiv: 2508.20079. To appear. 4. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Testing convex truncation”. <i>Mathematical Statistics and Learning</i> (2025). DOI: 10.4171/MSL/50. 5. Quentin Dubroff, Shivam Nadimpalli, and Bhargav Narayanan. “A Counterexample to a Directed KKL Inequality”. <i>Electronic Journal of Combinatorics</i> (2024). DOI: 10.37236/11828. 6. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Quantitative correlation inequalities via extremal power series”. <i>Probability Theory and Related Fields</i> (2022). DOI: 10.1007/s00440-022-01120-5. <i>Conference Publications</i> 1. Anindya De, Shivam Nadimpalli, Ryan O’Donnell, and Rocco A. Servedio. “Sparsifying Suprema of Gaussian Processes”. <i>Proceedings of the 56th ACM Symposium on Theory of Computing (STOC)</i>. 2026. DOI: 10.1145/3798129.3800761. 2. Yiqiao Bao, Anindya De, Shivam Nadimpalli, Rocco A Servedio, and Nathan White. “Testing noisy low-degree polynomials for sparsity”. <i>Proceedings of the 56th ACM Symposium on Theory of Computing (STOC)</i>. 2026. DOI: 10.1145/3798129.3800794. 3. Daeho Lee, Shivam Nadimpalli, Mingda Qiao, and Ronitt Rubinfeld. “Testing Unate Distributions”. <i>Proceedings of the 30th International Conference on Randomization and Computation (RANDOM)</i>. 2026. arXiv: 2607.01573. To appear.	

4. Renato Ferreira Pinto Jr, Cassandra Marcussen, Elchanan Mossel, and Shivam Nadimpalli. “Learning and Testing Convex Functions”. *Proceedings of the 30th International Conference on Randomization and Computation (RANDOM)*. 2026. arXiv: [2511.11498](#). To appear.
5. Xi Chen, Anindya De, Yizhi Huang, Shivam Nadimpalli, Rocco A. Servedio, and Tianqi Yang. “Sublinear-query relative-error testing of halfspaces”. *Proceedings of the International Colloquium on Automata, Languages, and Programming (ICALP)*. 2026. DOI: [10.4230/LIPIcs.ICALP.2026.61](#).
6. Xi Chen, Anindya De, Yizhi Huang, Shivam Nadimpalli, Rocco A. Servedio, and Tianqi Yang. “Halfspaces are hard to test with relative error”. *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2026. DOI: [10.1137/1.9781611978971.21](#).
7. Josh Alman, Shivam Nadimpalli, Shyamal Patel, and Rocco A. Servedio. “Faster exact learning of k -term DNFs with membership and equivalence queries”. *Proceedings of the 66th IEEE Symposium on Foundations of Computer Science (FOCS)*. 2025. DOI: [10.1109/FOCS63196.2025.00051](#).
8. Josh Alman, Shivam Nadimpalli, Shyamal Patel, and Rocco A. Servedio. “DNF Learning via Locally Mixing Random Walks”. *Proceedings of the 57th ACM Symposium on Theory of Computing (STOC)*. 2025. DOI: [10.1145/3717823.3718262](#).
9. Xi Chen, Anindya De, Shivam Nadimpalli, Rocco A. Servedio, and Erik Waingarten. “Lower Bounds for Convexity Testing”. *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2025. DOI: [10.1137/1.9781611978322.13](#).
10. Xi Chen, Anindya De, Yizhi Huang, Yuhao Li, Shivam Nadimpalli, Rocco A. Servedio, and Tianqi Yang. “Relative-error monotonicity testing”. *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2025. DOI: [10.1137/1.9781611978322.11](#).
11. Xi Chen, Shivam Nadimpalli, Tim Randolph, Rocco A. Servedio, and Or Zamir. “Testing Sumsets is Hard”. *Proceedings of the European Symposium on Algorithms (ESA)*. 2025. DOI: [10.4230/LIPIcs.ESA.2025.14](#).
12. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Gaussian Approximation of Convex Sets by Intersections of Halfspaces”. *Proceedings of the 65th IEEE Symposium on Foundations of Computer Science (FOCS)*. 2024. DOI: [10.1109/FOCS61266.2024.00115](#). Under submission to *Annals of Applied Probability*.
13. Anindya De, Huan Li, Shivam Nadimpalli, and Rocco A. Servedio. “Detecting Low-Degree Truncation”. *Proceedings of the 56th ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: [10.1145/3618260.3649633](#).
14. Shivam Nadimpalli and Shyamal Patel. “Optimal Non-Adaptive Tolerant Junta Testing via Local Estimators”. *Proceedings of the 56th ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: [10.1145/3618260.364968](#).
15. Shivam Nadimpalli, Natalie Parham, Francisca Vasconcelos, and Henry Yuen. “On the Pauli Spectrum of QAC^0 ”. *Proceedings of the 56th ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: [10.1145/3618260.3649662](#). Presented at *Quantum Information Processing (QIP)*.
16. Xi Chen, Anindya De, Yuhao Li, Shivam Nadimpalli, and Rocco A. Servedio. “Testing Intersecting and Union-Closed Families”. *Proceedings of the 15th Innovations in Theoretical Computer Science Conference (ITCS)*. LIPIcs. 2024. DOI: [10.4230/LIPIcs.ITCS.2024.33](#).

17. Xi Chen, Anindya De, Yuhao Li, Shivam Nadimpalli, and Rocco A. Servedio. “Mildly exponential lower bounds on tolerant testers for monotonicity, unateness, and juntas”. *Proceedings of the 2024 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2024. DOI: [10.1137/1.9781611977912.151](https://doi.org/10.1137/1.9781611977912.151).
18. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Testing Convex Truncation”. *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2023. DOI: [10.1137/1.9781611977554.ch155](https://doi.org/10.1137/1.9781611977554.ch155).
19. Thomas Chen, Shivam Nadimpalli, and Henry Yuen. “Testing and Learning Quantum Juntas Nearly Optimally”. *Proceedings of the 2023 ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2023. DOI: [10.1137/1.9781611977554.ch43](https://doi.org/10.1137/1.9781611977554.ch43). Presented at *Quantum Information Processing (QIP)*.
20. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Convex Influences”. *Proceedings of the 13th Innovations in Theoretical Computer Science Conference (ITCS)*. LIPIcs. 2022. DOI: [10.4230/LIPIcs.ITCS.2022.53](https://doi.org/10.4230/LIPIcs.ITCS.2022.53).
21. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Approximating Sumset Size”. *Proceedings of the 2022 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2022. DOI: [10.1137/1.9781611977073.94](https://doi.org/10.1137/1.9781611977073.94).
22. Anindya De, Shivam Nadimpalli, and Rocco A. Servedio. “Quantitative Correlation Inequalities via Semigroup Interpolation”. *Proceedings of the 12th Innovations in Theoretical Computer Science Conference (ITCS)*. LIPIcs. 2021. DOI: [10.4230/LIPIcs.ITCS.2021.69](https://doi.org/10.4230/LIPIcs.ITCS.2021.69). Invited to *Geometric Aspects of Functional Analysis (Seminar Notes)*.

Manuscripts & Preprints

23. Shivam Nadimpalli. “Optimal Sparsification of Gaussian Processes”. 2026. arXiv: [2606.19763](https://arxiv.org/abs/2606.19763).
24. Shivam Nadimpalli, Mingda Qiao, and Ronitt Rubinfeld. “No Price Tags? No Problem: Query Strategies for Unpriced Information”. 2025. arXiv: [2511.06170](https://arxiv.org/abs/2511.06170).
25. Xi Chen, Anindya De, Yizhi Huang, Shivam Nadimpalli, Rocco A. Servedio, and Tianqi Yang. “Model-agnostic super-resolution in high dimensions”. 2025. arXiv: [2511.07846](https://arxiv.org/abs/2511.07846).

ADVISING

Undergraduate Students

- Daeho Lee (MIT) Summer 2024 to Present
- Caleb Pascale (MIT) Summer 2024
- Thomas Chen (Columbia → Stanford CS PhD) Spring 2022

INVITED TALKS

“Optimal Sparsification of Gaussian Processes” [[Nad26](#)]

- Princeton Theory Lunch October 2026
- Workshop on Local Algorithms (WOLA) 2026 August 2026

“On Passive Models of Property Testing”

- Testing in the Modern World (STOC 2026 Workshop) June 2026

“Sparsifying Intersections of Halfspaces” [[De+26](#)]

- UMass Amherst Theory Seminar April 2026
- Cornell Theory Seminar November 2025
- Northwestern CS Theory Seminar October 2025
- University of Chicago Combinatorics and TCS Seminar October 2025
- Brown University Probability Seminar October 2025

	• NYU Theory Seminar	March 2025
	• Harvard/MIT/MSR Reading Group	February 2025
	“Mode-agnostic super-resolution” [Che+25b]	
	• Chicago Junior Theorists Workshop 2025	December 2025
	“Testing Convexity” [Che+25c]	
	• Simons Institute (Sublinear Algorithms Reunion Workshop)	July 2025
	“Gaussian Polytope Approximation” [DNS24]	
	• University of Washington Theory Seminar	April 2026
	• Harvard Theory Seminar	April 2026
	• CMU Theory Lunch	February 2025
	• Boston University Theory Seminar	February 2025
	• MIT Algorithms and Complexity Seminar	November 2024
	• Hausdorff Institute of Mathematics (Analysis & TCS Workshop)	November 2024
	• UMass Amherst Theory Seminar	October 2024
	• Brown University Theory Seminar	September 2024
	• Simons Institute (Analysis & TCS Reunion Workshop)	July 2024
	• University of Michigan CS Theory Seminar	December 2023
	• Stanford University CS Theory Seminar	November 2023
	“On the Pauli Spectrum of QAC^0 ” [Nad+24]	
	• University of Pennsylvania Theory Seminar	September 2023
	“Testing Convex Truncation” [DNS23]	
	• Simons Institute (Sublinear Algorithms Program Weekly Seminar)	June 2024
	• Northwestern University Theory Seminar	November 2023
	• Probability and Analysis Online Webinar	October 2023
	• UC Berkeley Theory Seminar	April 2023
	• DIMACS Theory of Computing Seminar	February 2023
	“Convex Influences and a Quantitative Gaussian Correlation Inequality” [DNS21; DNS22b]	
	• Rutgers Discrete Mathematics Seminar	April 2023
	• New York Colloquium on Algorithms and Complexity	December 2022
	“Approximating Sumset Size” [DNS22a]	
	• Stanford University CS Theory Seminar	March 2022
AWARDS	Rising Star in TCS (TCS for All)	2026
	Simons–Berkeley Research Fellowship	2024
	Departmental Service Award, Columbia University	2022
	Sigma Xi, Brown University	2019
	Senior Prize, Department of Computer Science, Brown University	2019
	High Honors, Budapest Semesters in Mathematics	2018
	Kishore Vigyan Protsahan Yojana Fellowship, Government of India	2014
	National Talent Search Scholar, Government of India	2011
TEACHING	<i>Instructor (MIT)</i>	
	• 18.03 Ordinary Differential Equations (Course Administrator)	Fall 2026
	• 18.821 Project Laboratory In Mathematics (with W. Zhang)	Spring 2026
	• 18.091 Introduction to Metric Spaces	IAP/Winter 2026
	• 18.031 System Functions and the Laplace Transform	IAP/Winter 2026

- 18.03 Ordinary Differential Equations (Course Administrator) Fall 2025
- 18.200 Principles of Applied Mathematics (Recitation Lead) Spring 2025
- 18.091 Introduction to Metric Spaces IAP/Winter 2025
- 18.031 System Functions and the Laplace Transform IAP/Winter 2025
- 6.1200 Math for Computer Science (with Z. Abel and F. T. Leighton) Fall 2024

Graduate Teaching Assistant (Columbia University)

- COMS 4236 Introduction to Computational Complexity Spring 2022
- COMS 4252 Introduction to Computational Learning Theory Spring 2021

High-School Mathematics Outreach

- Fun with Mathematical Inequalities (Columbia Science Honors Program) Fall 2021
- Introduction to the Theory of Computing (Alec Sun's Mathcamp) Summer 2020

SERVICE

- Co-Organizer for **TCS+** Fall 2024 to Present
- Co-Organizer for **Testing in the Modern World** (STOC 2026 Workshop) June 2026
- Co-Organizer for **WALDO 2025** April 2025
- Mentor for MIT Mathematics DRP January 2025
- Co-Organizer for Stochastic Calculus Reading Group, Simons Institute Summer 2023
- Co-Organizer for Columbia Theory Lunch 2021–2023
- Co-Organizer for Columbia Theory Student Seminar 2020–2022
- Mentor for Columbia Undergraduate Theory Seminar 2020–2022
- Webmaster for Columbia Theory Website 2019–2024

REVIEWING

FOCS, STOC, SODA, ITCS, CCC, COLT, TQC, RANDOM, QIP, ICALP
 Quantum, SIAM Journal on Computing, Electronic Journal of Combinatorics